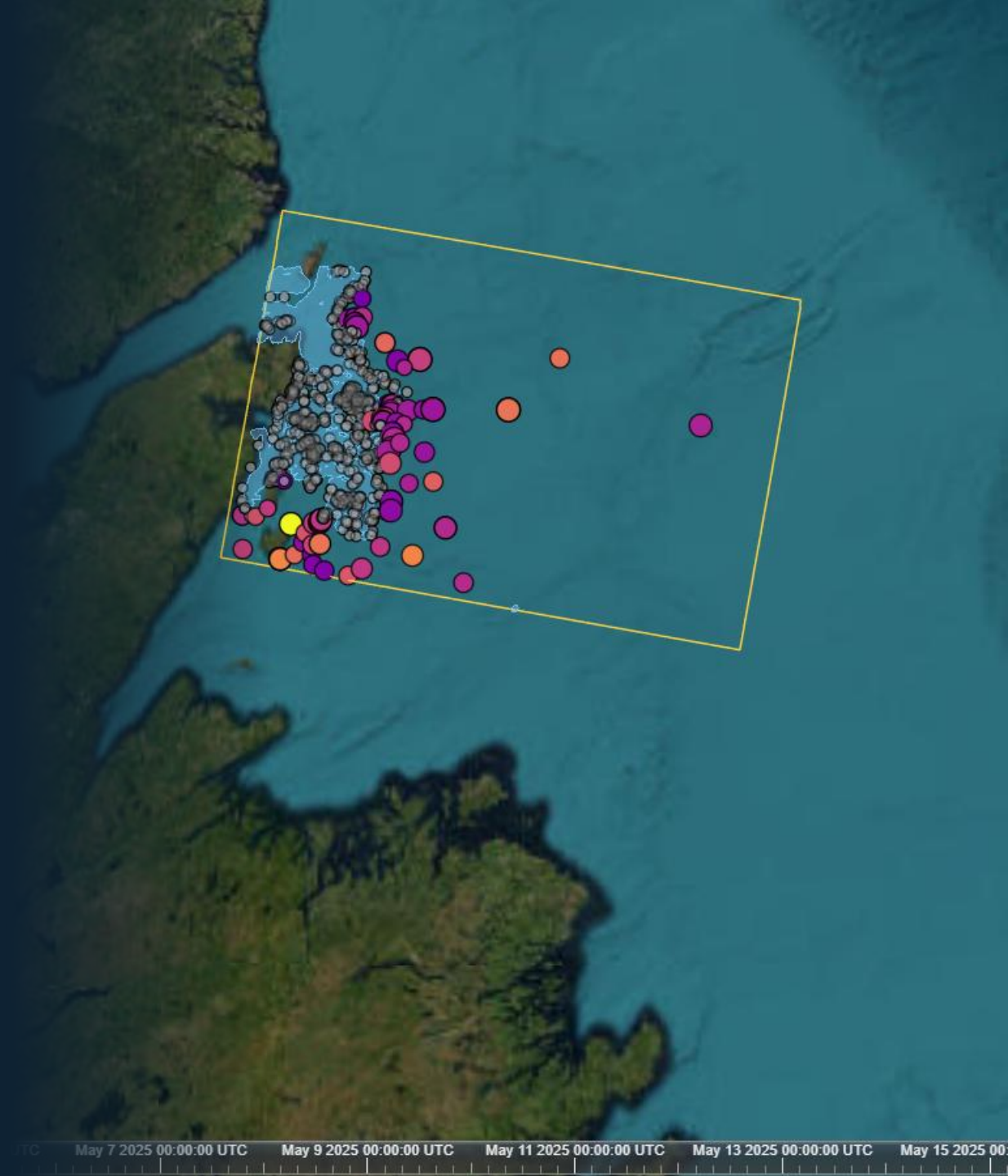
PROJECT BRIEF · SENTINEL-1 SAR

Iceberg Alley SAR

Detecting icebergs off Newfoundland in satellite radar with a classical detector and a foundation model, shown on a live interactive globe.

Natali Kovalski · October 2026



THE PROBLEM

Every spring, icebergs drift into busy water

01

The hazard

Greenland bergs ride the Labrador Current down "Iceberg Alley", past shipping lanes, fishing grounds and offshore platforms.

02

The gap

Aircraft reconnaissance is costly and limited by weather, and this coast is often fogged in.

03

The idea

Satellite radar sees through cloud and darkness. Sentinel-1 data is free and revisits every few days.

APPROACH

Six steps from raw scene to globe

01

Search

ASF search for HH+HV scenes over the AOI

02

Calibrate

HyP3 terrain correction: gamma0, linear power, 10 m

03

Mask

Land plus 500 m from OSM; pack ice from HV texture

04

Detect

CA-CFAR on HV, threshold set from the scene's speckle

05

Filter

Size limits, ice-strip rejection, HH co-pol check

06

Publish

GeoJSON and CZML, published to a live web viewer

Every parameter lives in **config.yaml**, with lengths in metres so tuning carries over between 10 m and 20 m pixels.

SAR IN ONE SLIDE

Four ideas behind the detector

Backscatter

Calm water reflects radar away and looks dark. An iceberg's facets send energy back and look bright.

Polarization

Sea clutter is far weaker in cross-pol (HV), so bergs stand out there. HH is the second check.

Speckle

Grainy noise in every SAR image. Its strength (ENL) is measured per scene and sets the threshold.

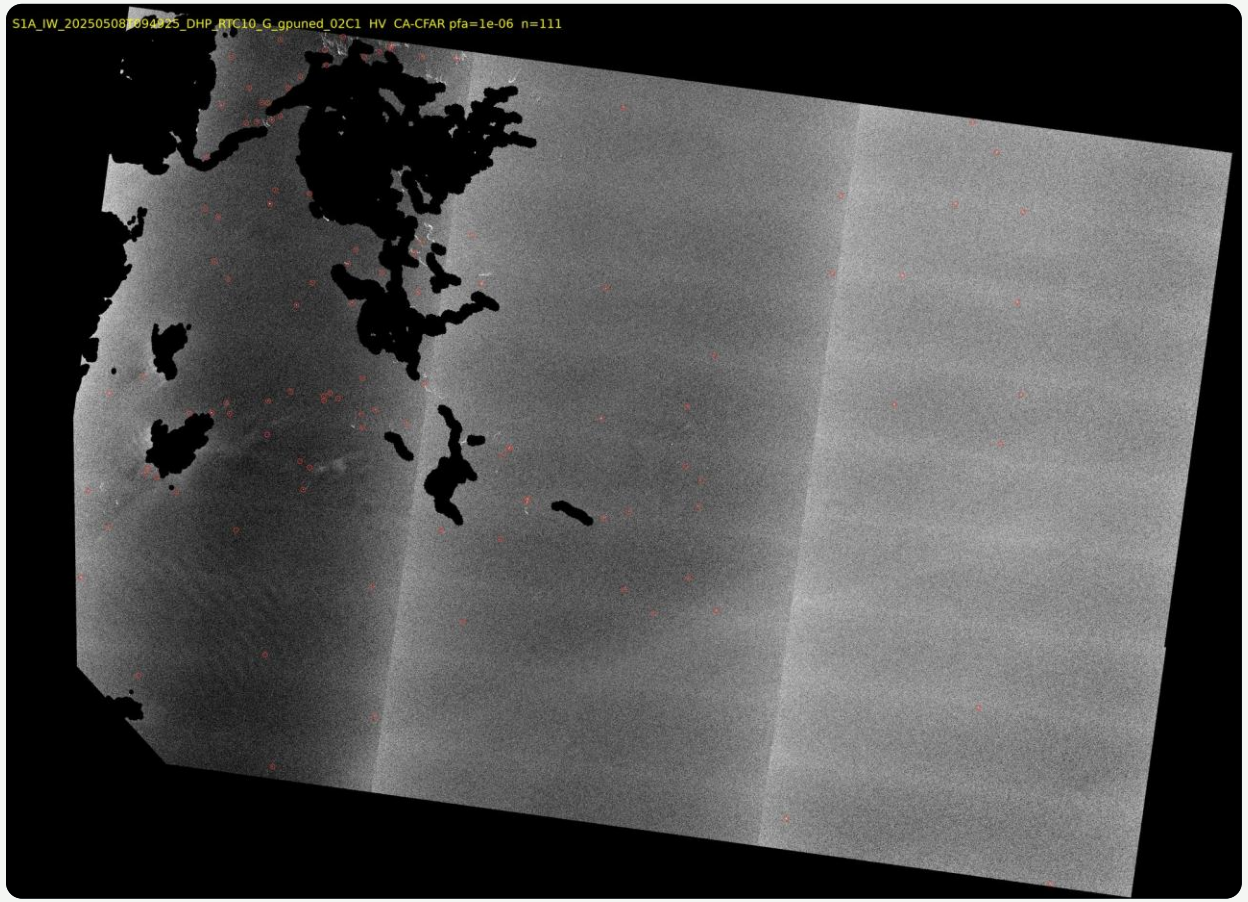
CFAR

Each pixel is compared with its local background, so false alarms stay rare in calm and rough seas.

DETECTION

CFAR on HV, then three filters

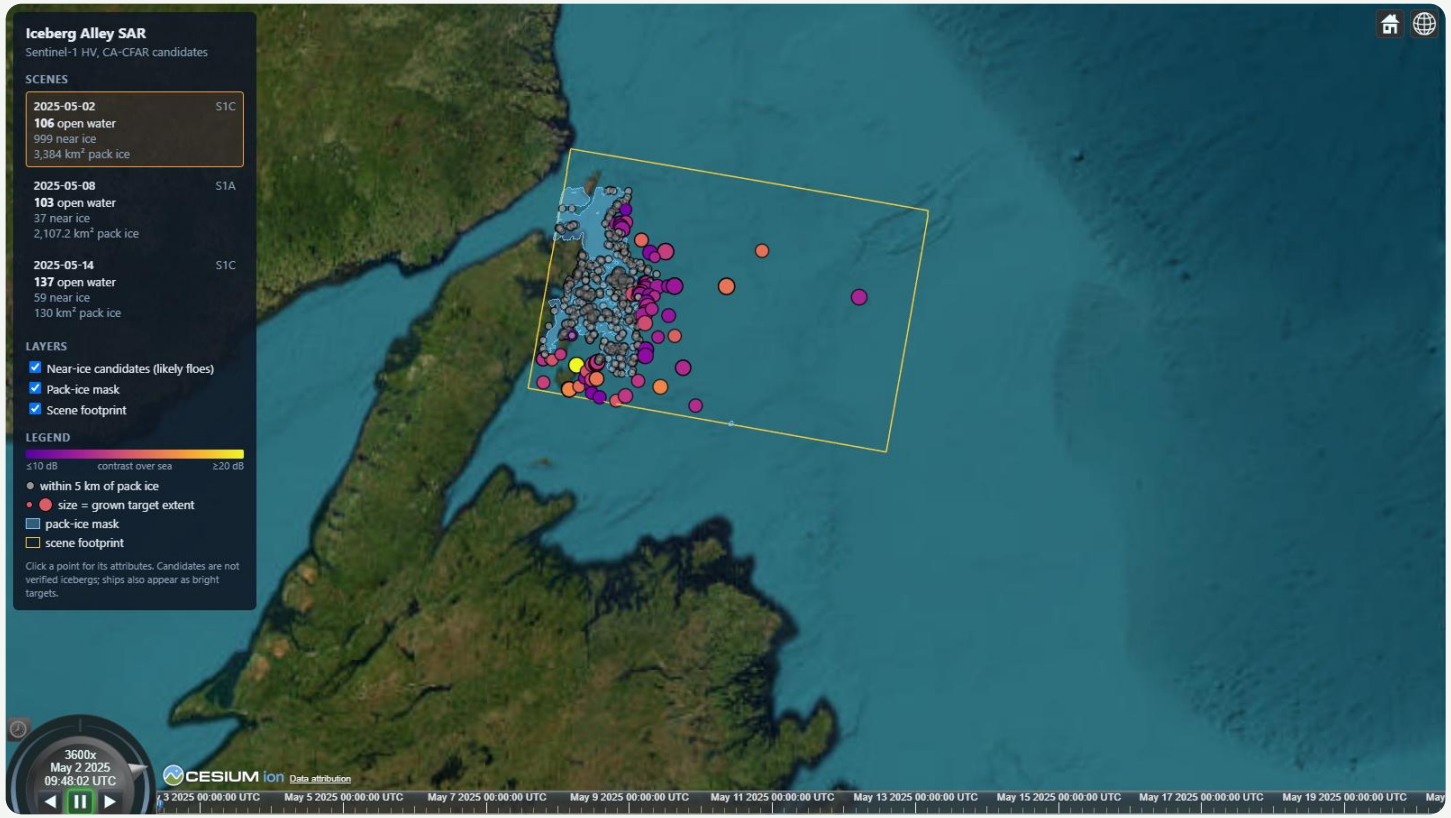
- **Threshold** from a 10^{-6} false-alarm rate, speckle measured per scene
- **Size:** 2 px to 0.8 km²
- **Ice strips:** reject bright structures longer than 300 m
- **HH check:** at least 6 dB above background in both polarizations



HV, 2025-05-08. Land and pack ice masked (black), 111 detections circled.

VIEWER

Every candidate on a globe, by date



- 32 passes, 2019–2026, live on GitHub Pages
- Scrub the timeline, click a target to see its attributes
- Orange dots: CA-CFAR. Cyan rings: THOR (toggle)
- Grey: within 5 km of pack ice, likely floes

HEADLINE RESULT

78%

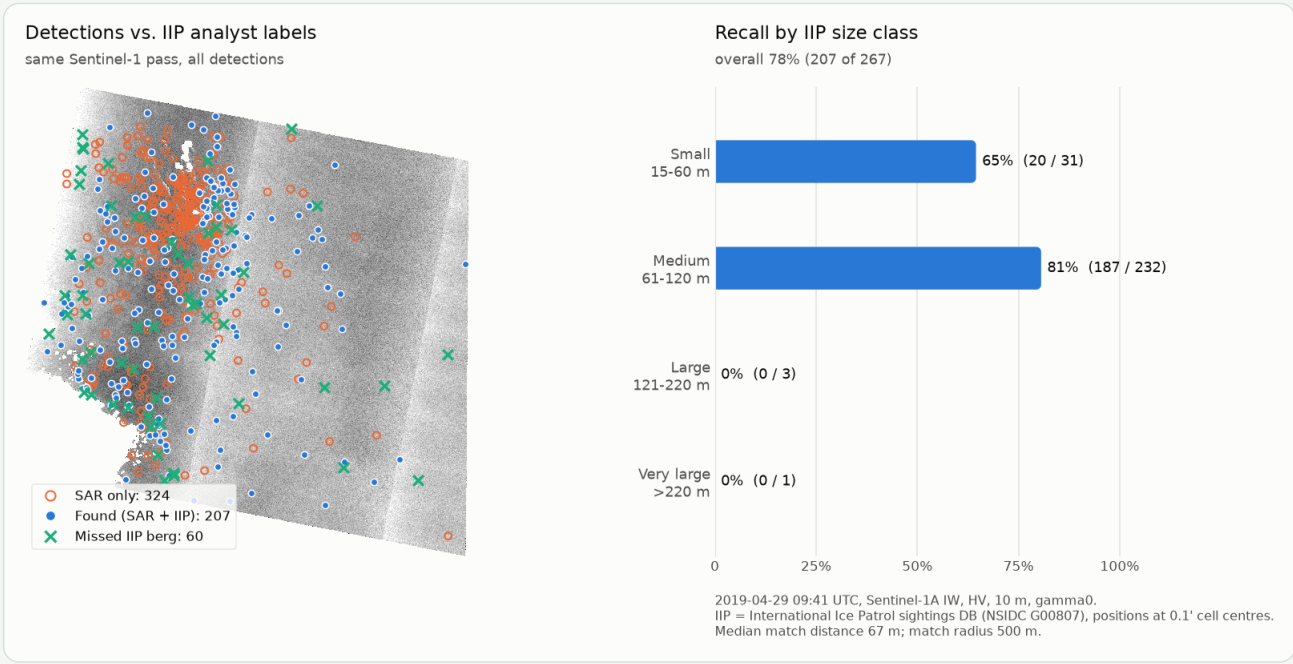
of the icebergs that International Ice Patrol analysts marked on the same Sentinel-1 pass were found: **207 of 267**.

2019-04-29 · median position difference 67 m · recall steady at 73–84% from the coast to 100 km offshore

RECALL BY SIZE

Medium bergs are found best

Berg size	Recall
Small, 15–60 m	65%
Medium, 61–120 m	81%
Large, over 120 m	0 of 4
All	78%



Detections vs. IIP analyst labels, 2019-04-29.

VALIDATION

Every ground-truth source has a catch

Source	What it gives	Outcome
NAIS iceberg chart	Counts per 1° square	Within 2–4×, but opposite errors cancel
IIP aircraft sightings	Positions 5–7 h after the pass	Inconclusive: drift unresolved
Sentinel-2 optical	Bright objects at 10 m	Loose sea ice in April, bergs inshore in June
IIP labels, same pass	Positions on the same image	Headline: 78% recall, no drift

Where truth comes hours later, the code estimates a common drift offset and checks it against chance. If the peak is weak, it reports "drift unresolved" instead of a score.

SECOND DETECTOR

A frozen foundation model, a tiny head

01

Frozen encoder

THOR Tiny (FM4CS, Apache 2.0), pretrained on Sentinel-1 sigma0. It takes HH/HV natively and is never fine-tuned.

02

769-parameter head

A linear layer on cached features, trained on IIP labels from 2019–2020 frames. All of April 2019 is held out.

03

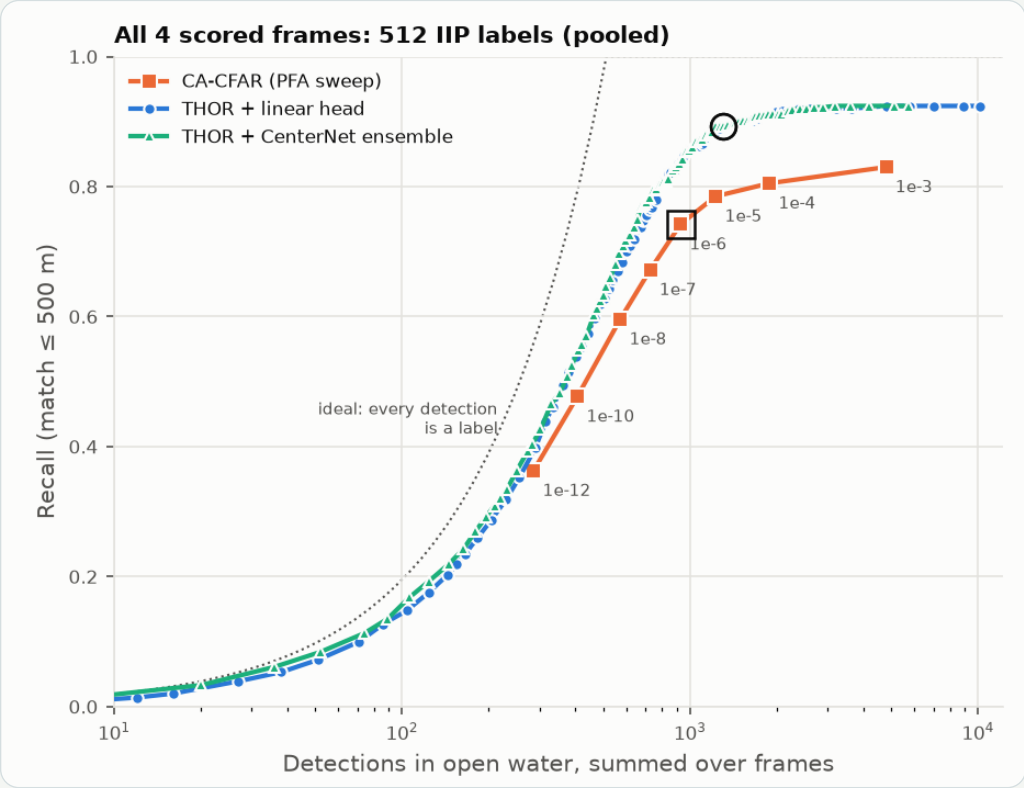
Same masks

Peaks must fall on the CFAR's masked raster. Output is the same GeoJSON, plus a score, so both run through one validation.

THOR VS. CA-CFAR

10 points more bergs at the same count

Recall	CFAR	THOR
4 frames, 512 bergs	0.74	0.84
2019-04-29 only	0.74	0.89
Medium, 61–120 m	0.76	0.87
Large, over 120 m	4 of 14	11 of 14
Small, 15–60 m	0.71	0.71



Recall vs. open-water detections, 4 held-out frames. Square: CFAR at PFA 10^{-6}.

LIMITS AND NEXT STEPS

What the numbers don't show yet

Limits

- **Precision unknown:** 324 of 531 detections on the test pass have no label (bergs, boats or clutter)
- **CFAR drops large bergs:** 4 of 14 found; its ice-strip rule rejects them
- **Rough seas** raise the HV background 3–4 dB
- **THOR needs the pipeline:** the pack-ice mask, and a GPU

Next

- **AIS ship positions** for the test passes, to measure precision
- **Shape-based strip test,** so the CFAR keeps large bergs
- **THOR at 4×4 patches** for small bergs
- **OS-CFAR or K-distribution** for heavy-tailed sea clutter

BUILD

One command, end to end

```
python -m iceberg_sar.cli run --start 2025-05-01 --end 2025-05-15
```

Processing

Python 3.11, NumPy, SciPy, rasterio, GeoPandas, PyTorch

Data

asf_search, HyP3 RTC, OSM land polygons, IIP sightings

Viewer

Vue 3, CesiumJS, CZML, deployed by GitHub Actions

Quality

153 pytest tests, all parameters in config.yaml

Re-runs reuse downloaded scenes and outputs, and a dry run shows the HyP3 credit cost first. Then publish and git push update the live viewer.

THANK YOU

Questions?

Natali Kovalski · github.com/natali-kovalski/iceberg-alley-sar

Live demo: natali-kovalski.github.io/iceberg-alley-sar. Contains modified Copernicus Sentinel data (2019–2026). Iceberg data: International Ice Patrol (NSIDC G00807) and the North American Ice Service. THOR weights: FM4CS (Apache 2.0). Land polygons © OpenStreetMap contributors.